# The Impact of Tax and Fee Reductions on the High-Quality Development of Non-Ferrous Metal Enterprises

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## Abstract

Drawing upon data from listed non-ferrous metal enterprises spanning 2013 to 2022, this study empirically examines how tax and fee reduction policies impact and drive mechanisms for fostering high-quality development within these entities. Findings indicate that reductions in taxes and fees significantly bolster such advancement. Moreover, a higher magnitude of reductions correlates with increased levels of R&D investment as well as expedited digital transformation—both pivotal factors in propelling superior growth among non-ferrous metal enterprises. Leveraging computer technology enables firms to enhance information acquisition capabilities, fortify internal communication channels, while utilizing cloud computing facilitates rapid comprehension of market dynamics. Consequently, insights gleaned from this research offer recommendations for refining tax relief measures furthering enhanced quality developments within non-ferrous metal enterprises.

## CCS Concepts

• **General and reference** → Document types; General literature.

## Keywords

Non-ferrous metal enterprises, Tax and fee reduction, R&D investment, Digital transformation, High-quality development of enterprises

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## 1 Introduction

The 20th National Congress of the Communist Party of China emphasized prioritizing high-quality development to comprehensively build a modern socialist country. The non-ferrous metal industry holds a critical position in China's industrialization and modernization, especially in recent years due to progress in emerging sectors like new energy, next-generation electronic information, and electric vehicles (Chen Xuesen, 2023)[1]. As an essential foundational material, non-ferrous metals are increasingly important. To advance high-quality development within non-ferrous metal enterprises, the government has enacted various policies and measures outlined in both the 'Non-Ferrous Industry Development Plan (2021–2035)' and 'Guiding Opinions on Promoting High-Quality Development of Non-Ferrous Metals Industry.' Improving production efficiency is vital for achieving high-quality development within these enterprises (Liu Siming et al., 2019)[2]. the key to promoting the development of high-quality enterprises lies in improving production efficiency. In the era of the digital economy, digital transformation is an effective approach to achieving this objective. The utilization of computer technology has become a crucial driving force behind digital transformation, enabling enterprises to efficiently manage business process data and optimize production processes and resource allocation. It's important to consider that implementing computer technology for digital transformation requires substantial capital investment. Fortunately, tax and fee reduction policies have increased available funds for non-ferrous metal enterprises, thereby facilitating greater levels of digital transformation within these organizations.Hence, tax and fee reduction, identified as a pivotal instrument for economic regulation and control (Gansu Provincial Taxation Society Research Group, 2024)[3], can effectively boost market vitality and the internal drive of enterprises.Mehmet (2011)[4] studied the relationship between tax reduction and fee reduction and tax burden through a dynamic combination model, and the results showed that reducing tax burden has a positive effect on improving corporate profits;Sedlacek and Sterk (2019) [5]found that reducing taxes and fees can make companies more willing to enter the market, thereby promoting the improvement of total factor productivity of enterprises. In recent years, there has been an increasing introduction of frequent policies aimed at reducing taxes and fees by the government, particularly after 2015. It is crucial for enterprises to capitalize on this opportunity to transform policy benefits into developmental momentum and advance their own high-quality development. Given this context, examining the impact and mechanism of tax reduction and fee reduction on the high-quality development of non-ferrous metal enterprises holds significant importance.

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**Table 1: High-quality development indicators of non-ferrous metal enterprises.**

| index | Indicator explanation |
|---|---|
| Scientific and technological innovation | R&D investment (R&D expenditure/total assets) |
| Green development | Environmental score in ESG |
| Social responsibility | Social Responsibility Score in ESG |
| Profitability | Operating margin, net profit margin on sales |
| Export scale | Overseas Earnings (Exports) |
| Talent team | Master's and doctoral ratio |

# 2 Research Design

## 2.1 Data sources

The paper utilizes a dataset comprising financial information from publicly listed non-ferrous metal companies between 2013 and 2022. The impact of China's tax and fee reduction measures, which began in 2012, gradually became evident in 2013. The dataset for the study is sourced from the Wind database, starting from 2013. To ensure data reliability, non-ferrous metal enterprises with insufficient disclosure, significant deficiencies, as well as those classified as ST and *ST between 2013 and 2022 were excluded after computer-assisted processing. Additionally, all variables were trimmed at the 1-99% level to mitigate the adverse effects of outliers. Ultimately, a panel dataset comprising of 5583 observations was obtained.

## 2.2 Variable selection

*2.2.1 Explanatory variables.* (1) Indicator selection

The academic community has not yet reached a consensus on the selection of indicators for measuring the high-quality development of enterprises. Some rely on standardized indicators such as labor productivity and total factor productivity, while others develop comprehensive index systems tailored to their specific industry characteristics. For non-ferrous metal enterprises, it is crucial to establish a comprehensive index system based on self-selected industry-specific indicators. This paper aligns with guidance on the high-quality development of the non-ferrous metals industry by selecting R&D investment, ESG environmental factors and social responsibility scores, operating profit margin and net sales profit margin, overseas income (export), and master's and doctoral ratio as measures for scientific and technological innovation, green development, social responsibility, profitability, export scale, and talent team construction.

(2) Evaluation methods for the high-quality development of non-ferrous metal enterprises

The current research on high-quality evaluation methods for non-ferrous metal enterprises is limited. Non-ferrous metal enterprises have few studies in this area. Drawing from the manufacturing industry's experience, TOPSIS and entropy methods are commonly utilized. While the TOPSIS method is more traditional, it fails to adequately reflect the importance and correlation between indicators. Therefore, this paper opts for the entropy method to address this deficiency in evaluation.

*2.2.2 Explanatory variables.* Drawing on the existing research by Li Yudong (2023) [6] and others, the total tax burden (ATR) is utilized as a surrogate indicator for assessing reductions in taxes and fees., the total tax burden is calculated as follows: total tax burden = (taxes paid - tax refund received) / operating income

*2.2.3 Intermediary variables.* (1) R&D investment (RD): The allocation of resources to research and development is a critical measure for evaluating an enterprise's dedication to innovation and progress. The magnitude of investment in research and development significantly impacts the high-quality growth of an enterprise.

(2) Digital transformation (DIG): In the age of the digital economy, embracing digital transformation signifies a strategic choice for enterprises to align with current trends, improve production efficiency, and leverage state-of-the-art information technology. The degree of digital transformation profoundly influences the high-quality advancement of non-ferrous metal enterprises.

*2.2.4 Control variables.* Based on the existing literature, this study incorporates enterprise age (AGE), enterprise size (SIZE), equity concentration (LARGE), capital density (CINT), operating income growth rate (RGR), and asset-liability ratio (LEV) as control variables..

# 3 Model building

Drawing on the analysis and research design of the influence mechanism in this paper, and building upon previous studies , **a computer modeling approach** was employed to construct a model for validating the research hypotheses presented in this paper. The following model was then established.

$$\begin{aligned}\mathrm{SCORE_{it}} &= \alpha + \beta_1\mathrm{ATR_{it}} + \beta_2\mathrm{SIZE_{it}} \\ &+\beta_3\mathrm{LARGE_{it}} + \beta_4\mathrm{CINT}+\beta_5 RGR + \beta_6 LEV + V_i + \varepsilon_{it}\end{aligned} \tag{1}$$

Among these, SCORE represents the independent variable indicating the degree of high-quality development in non-ferrous metal enterprises; ATR is the primary independent variable reflecting the overall tax burden, which demonstrates the practical implementation of tax and fee reduction policies at the enterprise level. V denotes individual effect variables controlling for specific enterprises, with subscripts i and t representing their respective values for non-ferrous metal enterprise i in year t.

# 4 Analysis of empirical results

## 4.1 Analysis of regression results

**With the assistance of STATA software and computer technology,** the samples were examined. The specific results can be found in Table 2. In column (1), the primary variable 'total tax burden' was independently regressed against the high-quality development of non-ferrous metal enterprises, with precise computer calculation

**Table 2: Impact of tax and fee reduction on the high-quality development of non-ferrous metal enterprises: benchmark regression.**

| Variable | (1) | (2) |
|---|---|---|
| | SCORE | SCORE |
| ATR | -0.179*** | -0.204*** |
| | (-4.02) | (-6.09) |
| SIZE | | 0.000*** |
| | | (16.36) |
| LARGE | | -0.000* |
| | | (-1.90) |
| CINT | | 0.000*** |
| | | (4.55) |
| RGR | | -0.000 |
| | | (-0.75) |
| LEV | | -0.001*** |
| | | (-9.29) |
| Corporate fixation | Control | Control |
| _cons | 0.133*** | 0.160*** |
| | (37.93) | (14.14) |
| N | 590 | 590 |

Note: ***, **, and * indicate significant at the 1%, 5%, and 10% levels, respectively; The t-value is enclosed in parentheses

and analysis confirming their relationship. The findings reveal a significant negative correlation at the 1% level between the total tax burden and high-quality development of non-ferrous metal enterprises, indicating an inverse relationship. Incorporating five control variables including enterprise size, equity concentration, capital intensity, operating income growth rate, and asset-liability ratio in column (2) resulted in regression outcomes that still demonstrate a substantial negative correlation at a 1% regression coefficient for the total tax burden when compared to column (1). Both regressions unequivocally illustrate evidence that policies aimed at reducing taxes and fees can significantly promote enterprise development by increasing available funds, alleviating financial constraints, boosting R&D investment, **leveraging computer technology for digital transformation acceleration** to reduce costs while greatly enhancing production efficiency.

## 4.2 Robustness test

*4.2.1 Shorten the sample interval.* In order to ensure the trustworthiness of the research findings, we **utilized computer data processing technology and analytical capabilities** to modify the original baseline regression interval from 2013-2022 to 2016-2022. The nationwide implementation in May 2016 replacing business tax with value-added tax aimed at achieving comprehensive coverage on goods and services while reducing enterprise's repetitive tax payments and overall burdens significantly impacting them. By using 2016 as a pivotal point, we can better capture the impact intensity from taxation and fee reduction policies. The re-regression model (1) yields results presented in Column (1) within Table 3 below. Despite altering our initial regression period, total tax burden coefficients remain notably negative at a significance level of one percent; furthermore, there were no significant changes observed in control variable significance—indicating robustness across our regression findings.

*4.2.2 Replace the explanatory variables.* The entropy method is employed to determine the dependent variable based on high-quality development guidance for non-ferrous metal enterprises. This approach is well-suited for the non-ferrous industry and is supported by robust justification. To ensure the reliability of regression results, **computer modeling and algorithm analysis** are utilized to replace total factor productivity calculated by the LP method as an indicator for measuring high-quality development in non-ferrous metals. Despite its lack of specificity, this approach, **with computer technology support**, offers greater universality and comprehensiveness while considering more factors than labor productivity. Furthermore, relying on accurate computer processing enables the LP method to overcome inaccuracies resulting from data deficiencies in OP method calculations; hence it is chosen to replace the explained variables in re-regression model (1). The results are presented in column (2) of Table 3. The regression coefficient for total tax burden remains significantly negative at a 1% level, with a larger coefficient indicating robustness of the regression result.

## 4.3 The impact mechanism of tax reduction and fee reduction on the high-quality development of non-ferrous metal enterprises

In order to examine the impact mechanism of tax and fee reduction on the high-quality development of non-ferrous metal enterprises, and to investigate whether such reductions can stimulate increased R&D investment and expedite the digital transformation of these

**Table 3: The impact of tax and fee reduction on the high-quality development of non-ferrous metal enterprises: robustness test.**

| Variable | (1) | (2) |
|---|---|---|
| | Shorten the sample interval | Replace the explanatory variable |
| | SCORE | TFP_LP |
| ATR | -0.248*** | -2.378*** |
| | (-5.91) | (-6.24) |
| SIZE | 0.000*** | 0.000*** |
| | (16.80) | (7.88) |
| LARGE | -0.000** | -0.018*** |
| | (-2.51) | (-6.69) |
| CINT | 0.000** | 0.000*** |
| | (2.55) | (8.21) |
| RGR | -0.000 | 0.002*** |
| | (-1.26) | (4.88) |
| LEV | -0.001*** | -0.008*** |
| | (-8.59) | (-4.29) |
| Corporate fixation | Control | Control |
| _cons | 0.175*** | 10.091*** |
| | (13.76) | (50.84) |
| N | 485 | 532 |

Note: ***, **, and * indicate significant at the 1%, 5%, and 10% levels, respectively; The t-value is enclosed in parentheses

enterprises, thereby enhancing production efficiency and promoting their high-quality development, this paper employs a gradual regression method with **computer technology**.

*4.3.1 The mediating effect test based on R&D investment.* The examination is conducted in two stages: firstly, to analyze the correlation **using computer technology** between total tax burden and R&D investment, with R&D investment as the independent variable and total tax burden as the primary explanatory variable, while maintaining constant control variables. In the second stage, **also utilizing computer technology**, regression analysis is performed with the high-quality development index of non-ferrous metal enterprises calculated by the entropy method as an additional independent variable alongside total tax burden and R&D investment as core explanatory variables to evaluate whether R&D investment has a mediating effect.

The first step is to verify the impact of tax reduction and fee reduction on enterprise R&D investment, and the measurement model is as follows:

$$\mathrm{RD_{it}} = \alpha_0 + \alpha_1 \mathrm{ATR}_{it} + \alpha_2 X_{\mathrm{it}} + V_i + \varepsilon_{it} \tag{2}$$

Among them, RD refers to the level of enterprise R&D investment, ATR refers to the total tax burden of enterprises, X refers to a series of control variables, and V refers to the variables that control the individual effect of enterprises.

In the second step, the model incorporates the total tax burden of enterprises and R&D input as the core explanatory variables, while taking high-quality development of non-ferrous metal enterprises as the explained variable to examine whether R&D input serves as an intermediary. The measurement model is:

$$SCORE_{it} = \alpha + \beta_1 ATR_{it} + \beta_2 RD_{it} + \beta_3 X_{it} + V_i + \varepsilon_{it} \tag{3}$$

Among them, the SCORE refers to the high-quality development level of non-ferrous metal enterprises, while the other variables remain in line with those in model (2).

The results of the intermediary mechanism analysis for R&D investment are detailed in Table 4. **Through precise computer technology calculations and data analysis**, it is evident that the regression coefficient of the overall tax burden in column (1) on R&D investment shows a significant negative correlation at a 1% level. This indicates that reductions in taxes and fees are inversely associated with R&D investment, suggesting that policies aimed at reducing taxes and fees can enhance enterprises' R&D investments and foster scientific and technological innovation. In column (2), the regression coefficient of the total tax burden and R&D investment on high-quality development within non-ferrous metal enterprises also exhibits a significantly negative correlation at a 1% level. Notably, this coefficient has decreased from its original value [as shown in column (3)] of 0.204 to 0.131, signifying that tax and fee reduction policies will augment R&D investments within non-ferrous metal enterprises. Elevating levels of R&D investment will bolster enterprise core competitiveness, enable mastery over cutting-edge information, and promote high-quality enterprise development. Therefore, **meticulous computer technology analysis reveals** that tax and fee reductions impact high-quality development within non-ferrous metal enterprises by stimulating research and development investments.

*4.3.2 The mediating effect test based on digital transformation.* The first stage involves evaluating the influence of tax and fee reductions on the digital transformation of enterprises, utilizing the subsequent measurement model:

$$\mathrm{DIG_{it}} = \alpha_0 + \alpha_1 ATR_{it} + \alpha_2 X_{it} + V_i + \varepsilon_{\mathrm{it}} \tag{4}$$

**Table 4: The impact mechanism of tax and fee reduction on the high-quality development of non-ferrous metal enterprises: the intermediary mechanism of R&D investment.**

| Variable | (1) | (2) | (3) |
|---|---|---|---|
| | RD | SCORE | SCORE |
| ATR | -0.028*** | -0.131*** | -0.204*** |
| | (-3.93) | (-4.63) | (-6.09) |
| SIZE | 0.000 | 0.000*** | 0.000*** |
| | (0.84) | (19.05) | (16.36) |
| LARGE | -0.000*** | -0.000 | -0.000* |
| | (-2.86) | (-0.37) | (-1.90) |
| CINT | 0.000*** | 0.000*** | 0.000*** |
| | (2.66) | (3.67) | (4.55) |
| RGR | -0.000 | -0.000 | -0.000 |
| | (-0.31) | (-0.69) | (-0.75) |
| LEV | -0.000*** | -0.001*** | -0.001*** |
| | (-9.86) | (-4.22) | (-9.29) |
| RD | | 2.580*** | |
| | | (15.17) | |
| Corporate fixation | Control | Control | Control |
| _cons | 0.034*** | 0.074*** | 0.160*** |
| | (13.78) | (6.69) | (14.14) |
| N | 590 | 590 | 590 |

Note: ***, **, and * indicate significant at the 1%, 5%, and 10% levels, respectively; The t-value is enclosed in parentheses

Among them, DIG represents the level of digital transformation in non-ferrous metal enterprises.

In the second step, total tax burden and enterprise digital transformation are considered as the core explanatory variables, while high-quality development of non-ferrous metal enterprises is taken as the dependent variable to examine whether digital transformation serves as an intermediate transmission mechanism. The measurement model is outlined as follows:

$$SCORE_{it} = \alpha + \beta_1 ATR_{it} + \beta_2 \mathrm{DIG}_{it} + \beta_3 X_{it} + V_i + \varepsilon_{\mathrm{it}} \quad (5)$$

Among them, the SCORE denotes the advanced development level of non-ferrous metal enterprises, while the remaining variables remain consistent with those in model (4).

**The intermediary mechanism facilitates enterprise digital transformation by harnessing the robust capabilities of computer technology and advanced algorithmic modeling methods. Professional statistical analysis software and efficient data processing algorithms are utilized to present the test results in the** 5. The regression coefficient of total tax burden on enterprise digital transformation is notably negative at a 1% level, as indicated in Table 1 below. A negative correlation between total tax burden and enterprise digital transformation suggests that reducing taxes and fees can expedite the digital transformation of non-ferrous metal enterprises. When regressing total tax burden and enterprise digital transformation to the core explanatory variables for high-quality development levels of enterprises, the regression coefficient for total tax burden remains significantly negative at a 1% level, as shown in Table 2 below. Furthermore, this coefficient has decreased from its original value of 0.204 to its current value of 0.141, indicating that enterprise digital transformation plays an intermediate role. **Enterprises effectively utilize computer technology** to enhance information acquisition, strengthen internal information communication, and employ cloud computing to swiftly comprehend market conditions. Policies aimed at reducing taxes and fees can accelerate enterprise digital transformation, reduce costs for enterprises, greatly enhance production efficiency, thereby promoting high-quality development within non-ferrous metal enterprises.

## 5 Conclusion

Drawing upon data from publicly traded companies in the non-ferrous industry spanning from 2013 to 2022, this study empirically examines how reductions in taxes and fees impact on fostering high-quality growth within non-ferrous metal firms. It reveals that enhancements in these reductions can elevate levels of R&D investment **while also enhancing computer applications** to expedite enterprise digital transformations via intermediary mechanism testing. The primary findings are twofold: Firstly, there exists a significant negative correlation between overall taxation burdens borne by non-ferrous metal firms with their pursuit for high-quality growth; such burdens impede their progress while reductions in taxes & fees bolster internal capital resources available for innovation & R&D efforts alongside scaling up operations thereby facilitating enterprise transformations & optimizations. Secondly, policies pertaining to tax & fee cuts serve as an intermediary transmission mechanism channeling investments into R&D activities & digital transformations aimed at propelling high-quality developments within non-ferrous metal firms.

The study furnishes a theoretical framework and empirical substantiation for the tax and fee reduction policy, yielding substantial

**Table 5: The impact mechanism of tax and fee reduction on the high-quality development of non-ferrous metal enterprises: the intermediary mechanism of digital transformation.**

| Variable | (1)<br>lnDIG | (2)<br>SCORE | (3)<br>SCORE |
|---|---|---|---|
| ATR | -3.054*** | -0.141*** | -0.204*** |
| | (-4.19) | (-4.67) | (-6.09) |
| SIZE | 0.000** | 0.000*** | 0.000*** |
| | (2.02) | (16.91) | (16.36) |
| LARGE | -0.031*** | -0.001*** | -0.000* |
| | (-6.46) | (-6.14) | (-1.90) |
| CINT | 0.000*** | 0.000 | 0.000*** |
| | (5.17) | (1.59) | (4.55) |
| RGR | 0.000 | -0.000 | -0.000 |
| | (0.07) | (-1.30) | (-0.75) |
| LEV | -0.014*** | -0.001*** | -0.001*** |
| | (-3.90) | (-8.45) | (-9.29) |
| lnDIG | | 0.007*** | |
| | | (3.75) | |
| Corporate fixation | Control | Control | Control |
| _cons | 4.538*** | 0.203*** | 0.160*** |
| | (12.57) | (11.94) | (14.14) |
| N | 530 | 530 | 590 |

Note: ***, **, and * indicate significant at the 1%, 5%, and 10% levels, respectively; The t-value is enclosed in parentheses

practical ramifications. It emphasizes the need to prioritize income tax reduction, particularly focusing on lowering corporate income tax rates to directly impact corporate profits in profit distribution. Furthermore, it suggests implementing more targeted and inclusive preferential tax policies to support enterprise development, especially for private enterprises that are more responsive to such policies. Additionally, it highlights the substantial opportunities and challenges faced by the non-ferrous industry in light of carbon peaking and neutrality, emphasizing the importance of increased investment in research & development (R&D), accelerated green technology innovation, digital transformation, energy conservation, emission reduction, and carbon mitigation. The aim is to provide ample funds for non-ferrous metal enterprises while supporting their green digital transformation through tax and fee reductions.

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